1. Full Length Numerical Ability with Video Solution- Level -Medium — Solutions & Explanations

🎯 Infosys Systems Engineer (SE) | 📅 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

Total Questions: 20 | **Total Marks:** 29

Section: Numeriacal ability

Problem 1 • 720 litres of a mixture contain milk and water only in the ratio of x : 3 respectively. If 30 litres water is added into it, then the ratio of quantity of milk to that of water in the resultant mixture will become 3:2. Find the value of

○ PRACTICE VARIANT

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

720 litres of a mixture contain milk and water only in the ratio of x : 3 respectively. If 30 litres water is added into it, then the ratio of quantity of milk to that of water in the resultant mixture will become 3:2. Find the value of

A 10

✓ **Verified Answer:** Option () > 📹 **Video Solution:** https://www.youtube.com/watch?v=Lrg1kclilbY #

Problem 2 • A contractor agreed to construct a 6 km road in 200 days. He employed 140 persons for the work. After 60 days, he realized that only 1.5 km road has been completed. How many additional people would he need to employ to finish the work exactly on time?

○ PRACTICE VARIANT

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

A contractor agreed to construct a 6 km road in 200 days. He employed 140 persons for the work. After 60 days, he realized that only 1.5 km road has been completed. How many additional people would he need to employ to finish the work exactly on time?

A 40

✓ **Verified Answer:** Option () > 📹 **Video Solution:** https://www.youtube.com/watch?v=cdFZxrezzCo #

Problem 3 • A sum of Rs. 45500 is to be paid back in 3 equal annual installments. How much is each installment if the interest is compounded annually at 20% per annum ?

○ PRACTICE VARIANT

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard

A sum of Rs. 45500 is to be paid back in 3 equal annual installments. How much is each installment if the interest is compounded annually at 20% per annum ?

A 21600

✓ **Verified Answer:** Option () > 📹 **Video Solution:** https://www.youtube.com/watch?v=ReF1N-z2u-s #

Problem 4 • Note : Type ans in Min Only without unit

PRACTICE VARIANT

ID: Q4 Source: Infosys PYQ Level: Infosys SE Standard

Note : Type ans in Min Only without unit

A 72

✓ **Verified Answer:** Option () > 🎥 **Video Solution:** https://www.youtube.com/watch?v=j91psFOaZTY #

Problem 5 • Note : Ans in days

PRACTICE VARIANT

ID: Q5 Source: Infosys PYQ Level: Infosys SE Standard

Note : Ans in days

A 6

✓ **Verified Answer:** Option () > 🎥 **Video Solution:** https://www.youtube.com/watch?v=Ba4Sj5hiG5o #

Problem 6 • If a 10-digit number= 5432y1749x is divisible by 72, then what is the value of (5x-4y)?

PRACTICE VARIANT

ID: Q6 Source: Infosys PYQ Level: Infosys SE Standard

If a 10-digit number= 5432y1749x is divisible by 72, then what is the value of (5x-4y)?

A 14

✓ **Verified Answer:** Option () > 🎥 **Video Solution:** https://www.youtube.com/watch?v=vD8QfVM6C_g #

Problem 7 • **Options:**

PRACTICE VARIANT

ID: Q7 Source: Infosys PYQ Level: Infosys SE Standard

A 240000

✓ **Verified Answer:** Option () > 🎥 **Video Solution:** https://www.youtube.com/watch?v=XapZjxCNDLg #

Problem 8 • If x is the mean of the first 8 positive integers multiple of 9 and Y is the Median for the same , then find the value of 5X-3Y.

◯ PRACTICE VARIANT

ID: Q8 Source: Infosys PYQ Level: Infosys SE Standard

If x is the mean of the first 8 positive integers multiple of 9 and Y is the Median for the same , then find the value of 5X-3Y.

A 81

✓ **Verified Answer:** Option () > 🎥 **Video Solution:** https://www.youtube.com/watch?v=uePg-C-XGog #

Problem 9 • **Options:**

◯ PRACTICE VARIANT

ID: Q9 Source: Infosys PYQ Level: Infosys SE Standard

A 45000

✓ **Verified Answer:** Option () > 🎥 **Video Solution:** https://www.youtube.com/watch?v=MLCC0vdAJNw #

Problem 10 • X and Y enter into a business partnership with capital in the ratio 3:5. After 5 months, X adds 50% of his capital, while Y withdraws 60% of his capital. What is the share (in Rs lakhs) of X in the annual profit of Rs 10.26 lakhs?

◯ PRACTICE VARIANT

ID: Q10 Source: Infosys PYQ Level: Infosys SE Standard

X and Y enter into a business partnership with capital in the ratio 3:5. After 5 months, X adds 50% of his capital, while Y withdraws 60% of his capital. What is the share (in Rs lakhs) of X in the annual profit of Rs 10.26 lakhs?

A 5.89

B 4.56

D 6.24

✓ **Verified Answer:** Option (C) > 🎥 **Video Solution:** https://www.youtube.com/watch?v=IspIF698ufU #

Problem 11 • **Options:**

◯ PRACTICE VARIANT

ID: Q11 Source: Infosys PYQ Level: Infosys SE Standard

A - ✅ **(B) ** *(Correct Answer)*

C - (D)

✓ **Verified Answer:** Option (B) > 🎥 **Video Solution:** https://www.youtube.com/watch?v=SYLTvjnvJKY #

Problem 12 • **Options:**

PRACTICE VARIANT

ID: Q12 Source: Infosys PYQ Level: Infosys SE Standard

B - (C)

✓ **Verified Answer:** Option (A) > **Video Solution:** https://www.youtube.com/watch?v=djrfWjlxqeg #

Problem 13 • **Options:**

PRACTICE VARIANT

ID: Q13 Source: Infosys PYQ Level: Infosys SE Standard

A 35km/hr

B 40km/hr

D 45 km/hr

✓ **Verified Answer:** Option (C) > **Video Solution:** https://www.youtube.com/watch?v=VvcgwtTcP_U #

Problem 14 • **Options:**

PRACTICE VARIANT

ID: Q14 Source: Infosys PYQ Level: Infosys SE Standard

B 3,7,8

C 3,6,9

D 4,7,7

✓ **Verified Answer:** Option (A) > **Video Solution:** https://www.youtube.com/watch?v=uN3uu8Awy-0 #

Problem 15 • A shopkeeper earns a profit of 20% by selling an article by giving 25% discount on its marked price. If the cost price of the article increases by 10%, then how much discount percent should he give now on the same marked price to earn the same percentage of profit as earlier?

PRACTICE VARIANT

ID: Q15 Source: Infosys PYQ Level: Infosys SE Standard

A shopkeeper earns a profit of 20% by selling an article by giving 25% discount on its marked price. If the cost price of the article increases by 10%, then how much discount percent should he give now on the same marked price to earn the same percentage of profit as earlier?

B 16.5%

C 18%

D 15%

✓ **Verified Answer:** Option (A) > **Video Solution:** https://www.youtube.com/watch?v=9SHym9bcr8U #

Problem 16 • In the given questions, two quantities are given, one as 'Quantity 1' and another as 'Quantity 2'. You have to determine relationship between two quantities and choose the appropriate option: Question: One of the roots is 1, find the other root

◯ PRACTICE VARIANT

ID: Q16 Source: Infosys PYQ Level: Infosys SE Standard

In the given questions, two quantities are given, one as 'Quantity 1' and another as 'Quantity 2'. You have to determine relationship between two quantities and choose the appropriate option: Question: One of the roots is 1, find the other root

A Quantity 1 > Quantity 2

B Quantity 1 ≥ Quantity 2

D Quantity 2 ≥ Quantity 1

E Quantity 1 = Quantity 2 or Relation cannot be established

✓ **Verified Answer:** Option (C) > **Video Solution:** https://www.youtube.com/watch?v=hVISBoLKmVs #

Problem 17 • In a river, the ratio of the speed of stream and speed of a boat A in still water is 2 : 5. Again, in the same river ratio of the speed of stream and speed of another boat B in still water is 3 : 4. What is the ratio of the speeds of the first boat to the second boat in still water ?

◯ PRACTICE VARIANT

ID: Q17 Source: Infosys PYQ Level: Infosys SE Standard

In a river, the ratio of the speed of stream and speed of a boat A in still water is 2 : 5. Again, in the same river ratio of the speed of stream and speed of another boat B in still water is 3 : 4. What is the ratio of the speeds of the first boat to the second boat in still water ?

A 10 : 7

C 4 : 3

D 5 : 4

E 7 : 11

✓ **Verified Answer:** Option (B) > **Video Solution:** https://www.youtube.com/watch?v=sisAFLdcRVg #

Problem 18 • Find the area of blue portion.

◯ PRACTICE VARIANT

ID: Q18 Source: Infosys PYQ Level: Infosys SE Standard

Find the area of blue portion.

A 52

B 62

C 72

✓ **Verified Answer:** Option (D) > **Video Solution:** https://www.youtube.com/watch?v=L3kD_FndH5o #

Problem 19 • In the given questions, two quantities are given, one as 'Quantity 1' and another as 'Quantity 2'. You have to determine relationship between two quantities and choose the appropriate option: Question: Two dices are rolled simultaneously. Quantity 1: Probability that the sum of the numbers that appeared is a multiple of 5. Quantity 2: 1/6

○ PRACTICE VARIANT

ID: Q19 | Source: Infosys PYQ | Level: Infosys SE Standard

In the given questions, two quantities are given, one as 'Quantity 1' and another as 'Quantity 2'. You have to determine relationship between two quantities and choose the appropriate option: Question: Two dices are rolled simultaneously. Quantity 1: Probability that the sum of the numbers that appeared is a multiple of 5. Quantity 2: 1/6

B Quantity 1 ≥ Quantity 2

C Quantity 2 > Quantity 1

D Quantity 2 ≥ Quantity 1

E Quantity 1 = Quantity 2 or Relation cannot be established

✓ **Verified Answer:** Option (A) > 🎥 **Video Solution:** https://www.youtube.com/watch?v=OZWFA9Y3RM0 #

Problem 20 • What is the mean of the mode and median of the data given below? 8, 7, 5, 4, 9, 1, 3, 7, 6, 5, 2, 1, 6, 3, 9, 8, 4, 6

○ PRACTICE VARIANT

ID: Q20 | Source: Infosys PYQ | Level: Infosys SE Standard

What is the mean of the mode and median of the data given below? 8, 7, 5, 4, 9, 1, 3, 7, 6, 5, 2, 1, 6, 3, 9, 8, 4, 6

A 6.5

B 6.25

C 5.5

✓ **Verified Answer:** Option (D) > 🎥 **Video Solution:** https://www.youtube.com/watch?v=xiyzH-teu_E #